

occupation to each observation in our Census and CPS samples for 1960–1998. Using labor supply in FTEs as weights, we calculated means of each DOT measure for each occupation-industry-education-year cell. These means provide the primary outcome measures for our analysis. To attain compatibility between changing Census Industry Codes for 1960–1998, we use a crosswalk developed by Autor, Katz, and Krueger [1998] providing 140 consistent CIC industries spanning all sectors of the economy. This crosswalk includes all CIC industries and attains consistency by aggregating where necessary to the lowest common level of consistent industry definition among 1970, 1980, and 1990 CIC standards.

A.4. Composite Task Indicators from the DOT

To verify that our results are robust to plausible alternative selections of DOT variables, we formed composite indicators of our intended constructs using Principal Components Analysis. We chose a short list of alternative DOT variables that appeared relevant to each of our conceptual categories. These choices are Nonroutine Analytic Tasks: GED-MATH, GED-REASON, NUMBER, MVC; Nonroutine Interactive Tasks: DCP, GED-LANGUAGE, DEPL, VARCH; Routine Cognitive Tasks: STS, COLORDIS, REPCON, VOCPREP; Routine Manual Tasks: FINGDEX, MOTOR, FORM, MANUAL. Definitions and representative examples of these variables are found in U. S. Department of Labor [1972]. Using 1980 employment as weights, we performed principal components analysis for each set of variables to identify the linear combinations that maximized common variation subject to the constraint that the sum of squared vector weights is equal to one. In each case, we used the first principal component.

A.5. Calculating DOT Quantiles

To convert DOT measures into percentiles of the 1960 task distribution, we used DOT 1977 task measures paired to the 1960 Census to form an employment-weighted empirical cumulative distribution function of task input for each task measure across 1120 industry-education-gender cells: 140 industries, two gender, and four education levels (high school dropout, high school graduate, some college, and college graduate). We applied a small amount of interpolation to remove flat spots in the distribution. We inverted this empirical distribution and applied it to all DOT task measures in subsequent years by industry-education-gender

and year. These centile values are the unit of measure for all later analyses. A limitation of this methodology is that DOT values after 1960 may potentially lie outside the support of the 1960 distribution, leading to truncation. In practice, this issue affects at most 4 percent of the distribution of each task in any year. An earlier version of this paper [Autor, Levy, and Murnane, 2001] employed raw DOT scores rather than the percentile measures used here. Results were qualitatively identical.

A.6. Calculating within-Occupation Changes in DOT Task Measures: 1977–1991

To measure within-occupation changes in task content, we employed the 1991 Revised Fourth Edition of the *Dictionary of Occupational Titles*, available in electronic form from the National Academy of Sciences [1981]. Based on a study of select industries to determine which jobs had undergone the most significant occupational changes since the 1977 publication of the DOT fourth edition, DOT analysts introduced, revised, and eliminated occupational definitions for occupations that were observed to have most substantively changed between 1977 and 1991. A total of 2452 occupations were reviewed, updated, or were added; 646 nominal titles were revised; 136 titles were combined; and 75 were deleted. To provide a conservative measure of total task change, we assume that any occupation that was not revised in the 1991 DOT experienced no task change.

We used documentation from the North Carolina Employment Security Commission [1992a, 1992b] to construct a crosswalk between the 1991 DOT and 1997 DOT occupation codes. With this crosswalk, we applied DOT 1991 task variables to our 1971 CPS file, yielding a match rate of 99.9 percent. Of these matched occupations, 73 percent (weighted by employment) had been updated between 1977 and 1991 by DOT examiners. We then calculated DOT means by 1970 and 1980 COC occupations and gender using a procedure identical to that described in A.3, to obtain 1991 DOT task means for the 1977 occupations. The within-occupation variation that we exploit over 1977–1991 stems exclusively from reevaluation of occupational content by DOT examiners, rather than from changes in the relative size of DOT suboccupations within CIC occupations.

We assigned the 1977 DOT and 1991 DOT task measures to 470 consistent COC occupations in the 1984 and 1998 CPS samples (corresponding to the years of our CPS computer measure).

As with the industry data, we transformed the DOT task measures into percentiles of the base year distribution, in this case using occupational employment shares from 1984 to form employment weights.

A.7. Computer Usage Data from the Current Population Survey

Industry computer use frequencies were calculated from the October 1984 and 1997 School Enrollment Supplements to the Current Population Survey (CPS) as the weighted fraction of currently employed workers ages 18–65 who answered yes to the question, “Do you use a computer directly at work?” within consistent CIC industries. A computer is defined as a desktop terminal or PC with keyboard and monitor and does not include an electronic cash register or a handheld data device. To calculate these frequencies in 1984 and 1997, 61,712 and 56,247 observations were used, respectively.

A.8. Computer and Capital Investment Measures from the National Income and Products Accounts

We used data on industry capital stocks and flows of equipment, structures, and computers from the National Income and Product Accounts [U. S. Department of Commerce, 2002a, 2002b] for the years 1950–1998. All NIPA stock and investment variables are measured in real 1996 dollars. Investment variables measure cumulated real investment in the relevant asset over the prior ten years, except in 1998 where we use 1.25 times cumulated investment over the previous eight years. Deflation of NIPA measures is performed by the Bureau of Economic Analysis using primarily Producer Price Indexes (PPIs). PPIs for computer investment are based on quality adjustment, price linking, and hedonic regression methods. As denominators for capital/FTE and computer investment/FTE variables, we used Census and CPS samples to calculate FTEs by industry by year. Computer investment is calculated as the sum of investment in mainframe computers, personal computers, packaged and custom software, printers, terminals, storage devices, and other integrated devices. Structures and equipment variables are defined in the NIPA.

To match CPS and Census data to the NIPA, we used a crosswalk developed by Autor, Katz, and Krueger [1998] and revised for this analysis to accommodate small changes in the NIPA sector scheme made during the recent NIPA revision. The resulting aggregation of NIPA and CIC data contains 47 consis-

tent industries covering all industrial sectors excluding Government and Private Households, spanning 1960–1998. Of these 47, we exclude from our analysis agriculture and government dominated services (5 NIPA industries).

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